

Web scraping & HTML tables

Friday, June 12, 2026

i Today: Friday June 12

Before class

- **Perusall Assignment:** [DC §16.3](#): HTML tables & web scraping

Plan for today

- Why scraping, and the ethics of doing it responsibly
- A 5-minute tour of HTML
- `rvest`: `read_html()`, CSS selectors, `html_text2()`, `html_attr()`
- The easy win: `html_table()`
- Cleaning what you scrape (footnotes, times, dates)
- In-class activity

Helpful links

- [Syllabus](#) · [AI policy](#) · [Schedule](#) · [R4DS Ch. 24](#) · [DC Ch. 16](#) · [rvest reference](#)

Why scrape?

Every easy data format is alike

“Every easy data format is alike. Every difficult data format is difficult in its own way.” - with apologies to Leo Tolstoy

So we’ve worked with easily accessible datasets like `penguins`, `flights`, or CSVs that can be quickly imported with `read_csv()`. But a lot of data is only available on **web pages** that are formatted for human eyes rather than `read_csv()`.

Data scraping is gathering data from sources that aren't already in data-frame format; in our context, this is usually translating from the HTML a browser renders into a tidy data frame.

When to scrape (and when not to)

Before you scrape, **always check for a friendlier door**:

1. **Is there a download link?**
2. **Is there an API?**
 - Many sites (Spotify, the Census, GitHub) offer an [API](#) that returns clean JSON. Often a CRAN package already wraps it (`tidycensus`, `spotifyr`, `gh`).
3. **Only then, scrape the HTML.**

Scraping is flexible but fragile. If a website administrator redesigns its HTML, your code will no longer work. It's best to reserve web scraping for cases when no easier option is feasible.

Good web-scraping citizens

Ethical web-scraping

The rule of thumb from *R4DS*: you're usually on safe ground if the data is

- **public**: you didn't have to log in or click "I agree",
- **non-personal**: not names, emails, locations of identifiable people, and
- **factual**: facts and data aren't copyrightable (creative text and images are).

R4DS real-world cautionary example: 2016 researchers scraped ~70,000 **OkCupid** profiles. These are public, but deeply personal dating profiles. The researchers then released the profiles with no anonymization. It was widely condemned. *Public* does not mean *fair game* or *ethical*.

Be a polite guest

A scraper can fire off thousands of requests in seconds. To a small server that may look like an attack.

- **Throttle**: pause between requests. Don't hammer a server.
- **Cache**: never download the same page twice.
- **Identify yourself** and respect `robots.txt` and terms of service where you reasonably can.

The `polite` package automates all of this for you, pausing, catching, and checking `robots.txt` for you.

```
# install.packages("polite")
library(polite)
session <- bow("https://www.example.com") # introduce yourself
result  <- scrape(session) # politely fetch
```

Quick HTML tour

What a web page really is

HTML (HyperText Markup Language) is the language behind every web page. Strip away the styling and it's just nested text:

```
<html>
<head>
  <title>Page title</title>
</head>
<body>
  <h1 id='first'>A heading</h1>
  <p>Some text &amp; <b>some bold text.</b></p>
  <img src='myimg.png' width='100' height='100'>
</body>
</html>
```

The whole page is a **tree** of nested elements. Scraping works precisely *because* most pages with data have a consistent, repeating structure.

Anatomy of an element

- **Tags** come in pairs: `<p> ... </p>`. The content lives in between.
- **Attributes** are `name='value'` pairs inside the start tag.
- Two attributes matter most for scraping:
 - **id**: a *unique* label for one element (`id='first'`)
 - **class**: a *reusable* label shared by many elements (`class='weight'`)

Designers use `id` and `class` to *style* the page. We hijack them to *find our data*.

Example element

cute cats

- Start Tag (<a ... >): This opens the element and tells the browser where it begins.
- Attributes (href='/cats', class='link'): These live inside the start tag and provide extra information or settings (like the destination URL or CSS styling classes).
- Contents (cute cats): This is the actual text, image, or nested element that gets displayed and affected by the tag.
- End Tag (): This closes the element, telling the browser where it stops.

Useful tags to recognize

Tag	Meaning
<h1>–<h6>	headings
<p>	paragraph
/ /	unordered / ordered list / list item
	link (destination in the href attribute)
	image (source in the src attribute)
 / <i> / 	bold / italic / inline container
<table> <tr> <th> <td>	table, row, header cell, data cell

You don't need to memorize HTML, but it's useful to know enough to spot where your data is hiding. Google or the [MDN docs](#) are great resources to help fill in any gaps.

rvest

The rvest workflow

rvest (think “harvest”) is the tidyverse scraping package. It's installed with the tidyverse but loaded separately:

```
library(rvest)
```

Steps:

1. `read_html(url)`: download the page into R

2. `html_elements(css)`: select the pieces you want (a *CSS selector*)
3. `html_element(css)`: narrow to one piece per observation
4. `html_text2()` / `html_attr()` / `html_table()`: pull out the actual values

Practice

`minimal_html()` is a function that allows us to hand write HTML. We'll use it to learn, then use the same tools on real web pages.

```
html <- minimal_html("
  <h1>This is a heading</h1>
  <p id='first'>This is a paragraph</p>
  <p class='important'>This is an important paragraph</p>
")
html
```

```
{html_document}
<html>
[1] <head>\n<meta http-equiv="Content-Type" content="text/html; charset=UTF-
8 ...
[2] <body>\n<h1>This is a heading</h1>\n  <p id="first">This is a paragraph</ ...
```

Everything that follows works *identically* whether `html` came from `minimal_html()` or `read_html("https://...")`.

CSS selectors: three you'll use constantly

A **Cascading style sheets (CSS) selector** is a mini pattern language that provides a concise way of describing which HTML elements you want to extract from a web page.

Selector	Selects	Read it as
<code>p</code>	all <code><p></code> elements	“every paragraph”
<code>.important</code>	all elements with <code>class="important"</code>	“anything classed important” (dot = class)
<code>#first</code>	the element with <code>id="first"</code>	“the one with id first” (hash = id)
<code>p.important</code>	<code><p></code> elements that are <i>also</i> <code>class="important"</code>	combine them
<code>li b</code>	<code></code> <i>inside</i> an <code></code>	descendant (space)

```
html |> html_elements("p") # both paragraphs
```

```
{xml_nodeset (2)}  
[1] <p id="first">This is a paragraph</p>  
[2] <p class="important">This is an important paragraph</p>
```

```
html |> html_elements(".important") # just the important one
```

```
{xml_nodeset (1)}  
[1] <p class="important">This is an important paragraph</p>
```

```
html |> html_element("#first") # the one with id="first"
```

```
{html_node}  
<p id="first">
```

element vs. elements (this trips everyone up)

- `html_elements()` (plural): returns *all* matches. Length depends on the page.
- `html_element()` (singular): returns *one match per input*, filling NA when nothing matches.

```
html |> html_elements("b") # 0 matches -> empty
```

```
{xml_nodeset (0)}
```

```
html |> html_element("b") # 0 matches -> NA (length preserved!)
```

```
{xml_missing}  
<NA>
```

That NA-filling behavior keeps columns aligned.

Why the distinction matters

Below is a page of four StarWars characters, some of whom are missing a recorded weight:

```
droids <- minimal_html("
  <ul>
    <li><b>C-3PO</b> is a <i>droid</i> that weighs <span class='weight'>167 kg</span></li>
    <li><b>R2-D2</b> is a <i>droid</i> that weighs <span class='weight'>96 kg</span></li>
    <li><b>Yoda</b> weighs <span class='weight'>66 kg</span></li>
    <li><b>R4-P17</b> is a <i>droid</i></li>
  </ul>
")
```

Grab the weights *directly* and you get **3** values for **4** characters and no way to know which one is missing:

```
droids |> html_elements(".weight") |> html_text2() # length 3 - misaligned!
```

```
[1] "167 kg" "96 kg" "66 kg"
```

Rows then columns

Step 1. Use `html_elements()` to grab one element *per observation* (each `<li>` is one character):

```
characters <- droids |> html_elements("li")
length(characters) # 4 (one per character)
```

```
[1] 4
```

Step 2. Use `html_element()` *on each* to pull one value per observation. Missing weights become `NA`, so everything stays aligned:

```
tibble(
  name = characters |> html_element("b") |> html_text2(),
  species = characters |> html_element("i") |> html_text2(),
  weight = characters |> html_element(".weight") |> html_text2()
)
```

```
# A tibble: 4 x 3
  name    species weight
<chr> <chr>   <chr>
1 C-3PO  droid    167 kg
```

```
2 R2-D2  droid    96 kg
3 Yoda   <NA>     66 kg
4 R4-P17 droid    <NA>
```

Tip

`html_elements()` to find your *rows*, then `html_element()` to find each *column*.

Pulling out the values

Selecting elements gets you HTML nodes. Two functions turn nodes into data:

`html_text2()`: the visible text (handles whitespace like a browser; prefer it over `html_text()`):

```
droids |> html_elements("b") |> html_text2()
```

```
[1] "C-3P0" "R2-D2" "Yoda" "R4-P17"
```

`html_attr("name")`: the value of an attribute, e.g. a link's destination:

```
links <- minimal_html("
  <p><a href='https://en.wikipedia.org/wiki/Cat'>cats</a></p>
  <p><a href='https://en.wikipedia.org/wiki/Dog'>dogs</a></p>
")
links |> html_elements("a") |> html_attr("href")
```

```
[1] "https://en.wikipedia.org/wiki/Cat" "https://en.wikipedia.org/wiki/Dog"
```

Warning

`html_attr()` always returns a string. `html_attr("width")` gives "100", not 100. Follow up with `parse_number()` / `as.integer()`.

Tables

html_table()

If your data is already in an HTML `<table>`, you're in luck. `html_table()` reads it straight into a tibble, guessing column types as it goes.

```
tbl <- minimal_html("
  <table>
    <tr><th>x</th>   <th>y</th></tr>
    <tr><td>1.5</td> <td>2.7</td></tr>
    <tr><td>4.9</td> <td>1.3</td></tr>
    <tr><td>7.2</td> <td>8.1</td></tr>
  </table>
")
tbl |> html_element("table") |> html_table()
```

```
# A tibble: 3 x 2
      x     y
  <dbl> <dbl>
1   1.5   2.7
2   4.9   1.3
3   7.2   8.1
```

`x` and `y` came back as `<dbl>` automatically. (Turn that off with `convert = FALSE` when the auto-guess misfires.)

A real page: world record in the mile

The Wikipedia page on the [mile run world record progression](https://en.wikipedia.org/wiki/Mile_run_world_record_progression) holds a **dozen** tables. `html_elements("table")` grabs them all into a list:

```
url <- "https://en.wikipedia.org/wiki/Mile_run_world_record_progression"

tables <- url |>
  read_html() |>
  html_elements("table") |>
  html_table()

length(tables)      # 12 tables on the page
record <- tables[[4]] # the men's IAAF-era progression
```

```
# A tibble: 6 x 4
  Time   Athlete      Nationality Date
<chr>  <chr>          <chr>      <chr>
1 4:14.4 John Paul Jones United States 31 May 1913[6]
2 4:12.6 Norman Taber   United States 16 July 1915[6]
3 4:10.4 Paavo Nurmi     Finland      23 August 1923[6]
4 4:09.2 Jules Ladoumègue France        4 October 1931[6]
5 4:07.6 Jack Lovelock  New Zealand  15 July 1933[6]
6 4:06.8 Glenn Cunningham United States 16 June 1934[6]
```

Double brackets `[[4]]` because `tables` is a **list** of data frames. (Note: Wikipedia changes frequently, so table 4 today may differ tomorrow; always be sure to look before you trust an index.)

Cleaning scraped data

Scraped data is usually dirty

Look closely at what we got:

- Date is "31 May 1913[6]". Here, a footnote marker [6] is glued on.
- Time is "4:14.4", which is a string, not a number we can plot or subtract.

Scraped values are **always character strings**, often with human-formatting baggage. The next step after scraping is almost always cleaning:

- `str_remove()` / `gsub()` + **regular expressions** to strip junk
- `parse_number()` for currency, percentages, stray units
- `lubridate` (`dmy()`, `mdy()`, ...) for dates
- `as.duration(ms(...))` for times like "4:14.4"

Stripping footnotes with a regex

We want to delete a bracketed footnote like [6] at the end of the date. A **regular expression** describes that pattern:

```
record <- record |>
  mutate(Date = str_remove(Date, "\\[.\\]$"))
record$Date
```

```
[1] "31 May 1913"    "16 July 1915"    "23 August 1923" "4 October 1931"
[5] "15 July 1933"    "16 June 1934"
```

Decoding "\\[.\\]\$":

Piece	Means
\\[\\]	a <i>literal</i> [and] (escaped — brackets are special)
.	any single character (here, the footnote digit)
\$	anchored at the end of the string

We'll discuss regular expressions in more detail next week.

Parsing dates and times

Once we have a clean date, we can use the `lubridate` package. These are "4 October 1931"-style to **day-month-year** to `dmy()`:

```
record <- record |>
  mutate(
    Date = dmy(Date),
    seconds = as.numeric(as.duration(ms(Time))) # "4:14.4" -> 254.4 sec
  )
record |> select(Time, seconds, Date)
```

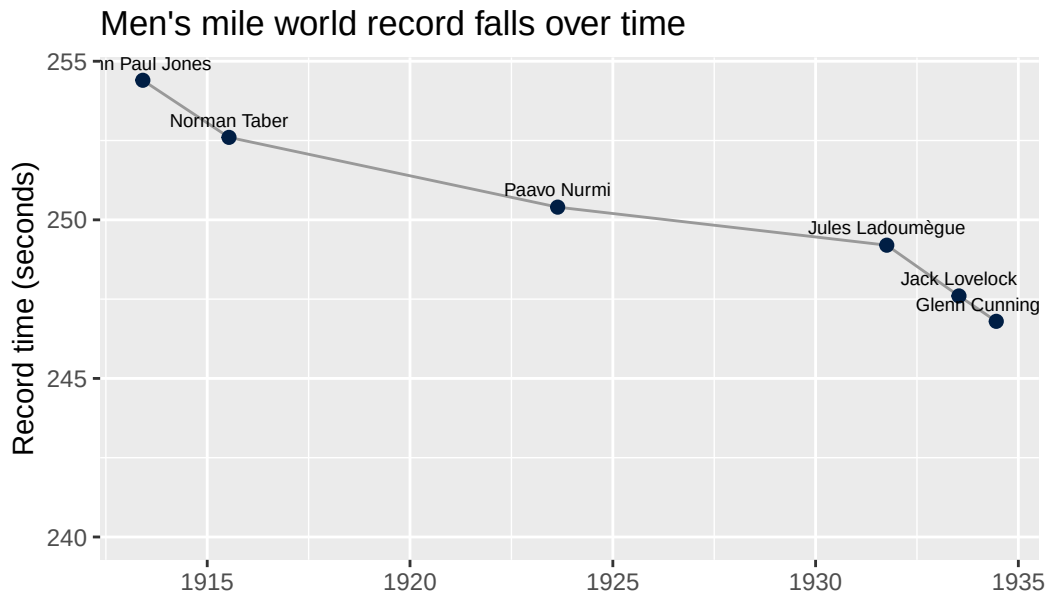
```
# A tibble: 6 x 3
  Time    seconds Date
<chr>    <dbl> <date>
1 4:14.4    254. 1913-05-31
2 4:12.6    253. 1915-07-16
3 4:10.4    250. 1923-08-23
4 4:09.2    249. 1931-10-04
5 4:07.6    248. 1933-07-15
6 4:06.8    247. 1934-06-16
```

`ms()` reads "minutes:seconds"; `as.duration()` turns it into a real time span; `as.numeric()` gives plain seconds we can plot.

Pay-off: a plot of progress

With clean `<date>` and numeric `seconds`, the record progression plots itself:

```
ggplot(record, aes(x = Date, y = seconds)) +  
  geom_line(color = "grey60") +  
  geom_point(size = 2, color = "#001E44") +  
  geom_text(aes(label = Athlete), vjust = -0.8, size = 2.6) +  
  labs(  
    title = "Men's mile world record falls over time",  
    x = NULL, y = "Record time (seconds)"  
  ) +  
  expand_limits(y = 240)
```



From scattered HTML to a finished plot — the whole import-and-clean pipeline in one slide.

Finding selectors

How do I know *which* selector?

This is the genuinely hard part — and it's normal to need trial and error. Three ways to discover the selector you need:

1. **Right-click then Inspect** in your browser. The developer tools highlight the exact HTML for whatever you clicked. Look at its `class` and `id`.
2. **SelectorGadget**: a point-and-click bookmarklet. Click what you *want* (turns green), click what you *don't* (turns red), and it writes the selector for you.
3. **Copy as selector**: in DevTools, right-click an element → Copy → Copy selector.

You want a selector that's **sensitive** (catches everything you want) *and* **specific** (catches nothing you don't).

Dynamic sites

Sometimes `html_elements()` returns *nothing* like what you see in the browser. Usually that means the page builds its content with **JavaScript** *after* loading.

`rvest` only downloads the raw HTML. It doesn't run JavaScript. This means you should:

- Check whether the data is in the *raw* HTML (View Source, not Inspect).
- If it's truly JS-rendered, you need a tool that drives a real browser: the `chromote` package runs Chrome in the background.
- Or, again: **look for an API first**. Dynamic pages are often fed by one.

Activity

In-class activity

Open `day19-activity.R` in RStudio and work through the parts in order.

- **Part 1**: HTML basics with `minimal_html()`: selectors, `element` vs `elements`, the alignment trap.
- **Part 2**: scrape a real Wikipedia table with `html_table()`, then clean dates and times and make a plot.
- **Part 3 (challenge)**: scrape *non-table* structured content (the `rvest` StarWars vignette) into a tidy tibble, and pull a link out of an attribute.

[Activity file](#)

Wrap-up & looking ahead

What we covered

- Scrape responsibly: public, non-personal, factual; throttle & cache
- HTML is a tree of tags, with `class` and `id` attributes
- `rvest`: `read_html()` → `html_elements()` → `html_element()` → `html_text2()` / `html_attr()`
- The golden rule: **rows with `html_elements()`, columns with `html_element()`**
- `html_table()` for the easy wins
- Clean scraped strings with `regex`, `parse_number()`, `lubridate`, `as.duration(ms())`

Upcoming

- **Mon Jun 15:** writing your own **functions** (Week 5)
- **HW 4** due Mon Jun 15
- **Project Checkpoint 1** due Sun Jun 14

Reach out

Stuck? Office hours, or email me (cgc5478@psu.edu) or Xinyue (xpw5228@psu.edu).

Reference

[Syllabus](#) · [AI policy](#) · [Schedule](#) · [R4DS Ch. 24](#) · [rvest reference](#) · [SelectorGadget](#)